APPLICATION NOTE

AN0006-EC NB-IoT HTTP Application Note_v1.1

EigenCOMM Wireless Microcontroller

Document Description

The purpose of present document is to help users quickly create http client and connect with http server.

EiGENCOMM
More than connection

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1. Foreword

1.1 Scope

The present document describes how to use HTTP AT command to implement user’s module connect to HTTP server as HTTP client. And access the HTTP server to complete various basic interactive operations.

1.2 HTTP protocol

The Hypertext Transfer Protocol(HTTP) is a stateless and application-level protocol. HTTP specification specifies how clients' request data will be constructed and sent to the server, and how the servers respond to these requests. The basic features of HTTP:

1．Client and server mode, basic authentication

2．Simple, The HTTP client sends a request to the server in the form of a request method, URI, and protocol version, followed by a MIME-like message containing request modifiers, client information, and possible body content over a TCP/IP connection. The HTTP server responds with a status line, including the message's protocol version and a success or error code.

3．Flexible: HTTP allows the transmission of any type of data object. The type be transmitted is marked by Content-Type.

4．HTTP1.1 uses persistent connections: one connection can handle multiple client requests.

5．Stateless: there is no link between two requests be successively carried out on the same connection.

HTTP rely on the TCP standard，When secure transmission is required, it is carried on the TLS or SSL protocol layer, that is, HTTPS. As shown below:

Application layer	HTTP	
		TLS、SSL
transport layer	TCP	
Network layer	IP	
Data link layer		

Figure 1：HTTP Overview

1.3 Brief introduction

The present document introduces the entire process of the HTTP AT command, and then lists the process of accessing the HTTP server.

2. Application

2.1 Interact with the HTTP server

The present document takes an example of accessing a weather query website to demonstrate the use of HTTP GET to obtain weather data for a city. The host of the weather website is http://api.openweathermap.org:80. The path to get the weather data of Shanghai is "/data/2.5/weather?q=shanghai&appid=c592e14137c3471fa9627b44f6649db4&mode=xml&units=metric". After sending a GET request, you will receive a response message from the website, as shown in the following figure.

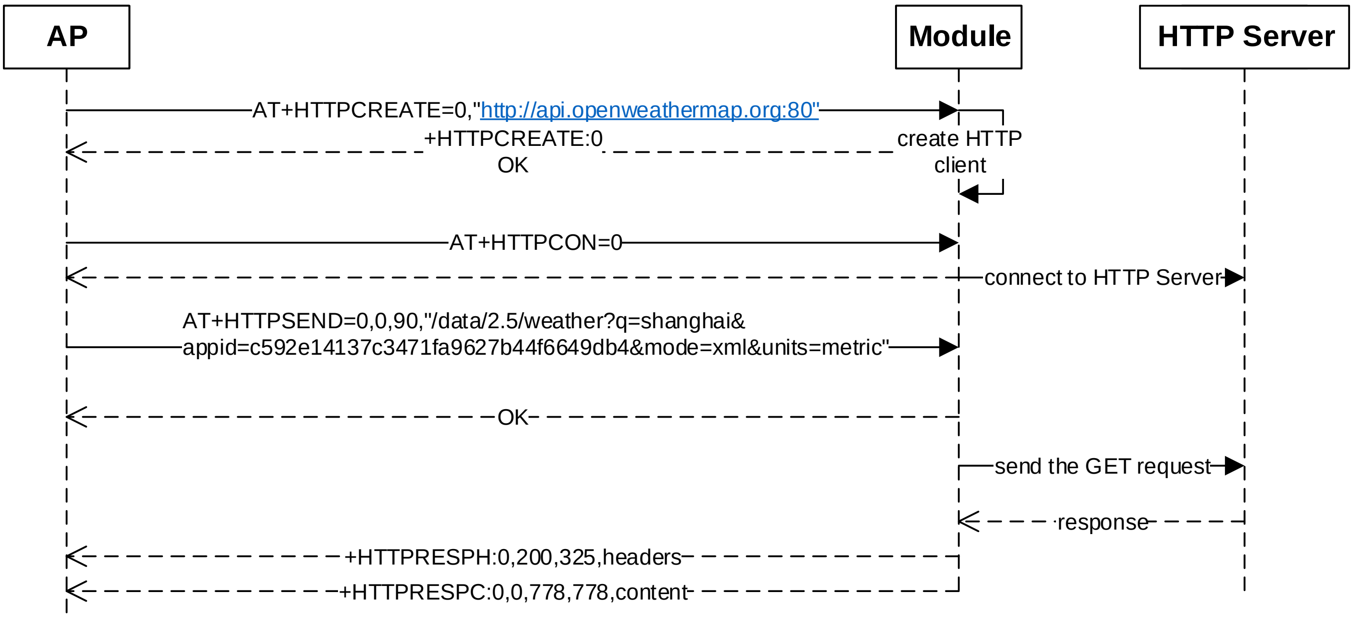

Figure 2：Interact with the HTTP server

The screenshot of the content of the website's reply on the serial port is as follows. The data is not processed and the customer can process it according to actual needs.

```
+HTTPRESPH:0,200,325,Server: openresty
Date: Sat, 02 Feb 2019 01:48:46 GMT
Content-Type: application/xml; charset=utf-8
Content-Length: 778
Connection: keep-alive
X-Cache-Key: /data/2.5/weather?mode=xml&q=shanghai&units=metric
Access-Control-Allow-Origin: *
Access-Control-Allow-Credentials: true
Access-Control-Allow-Methods: GET, POST

+HTTPRESPC:0,0,778,778,<?xml version="1.0" encoding="UTF-8"?>
<current><city id="1796236" name="Shanghai"><coord lon="121.49" lat="31.23"></coord><country>CN</
country><sun rise="2019-02-01T22:46:07" set="2019-02-02T09:29:35"></sun></city><temperature value="7.52"
min="7" max="8" unit="metric"></temperature><humidity value="81" unit="%"></humidity><pressure value="1020"
unit="hPa"></pressure><wind><speed value="6" name="Moderate breeze"></speed><gusts></gusts><direction
value="150" code="SSE" name="South-southeast"></direction></wind><clouds value="44" name="scattered
clouds"></clouds><visibility value="7000"></visibility><precipitation mode="no"></precipitation><weather
number="802" value="scattered clouds" icon="03d"></weather><lastupdate value="2019-02-02T01:00:00"></
lastupdate></current>
```

Figure 3：Screenshot of website's response

2.2 Detailed description of AT commands

2.2.1 AT+HTTPCREATE

Set command creates a http or https client instance. Configure host, server certification, etc.

Test command returns values supported as a compound value.

Note: only one instance and http was fully verified. https and multiple instances will be test later.

AT+HTTPCREATE

Set Command	Response

Set Command
`AT+HTTPCREATE=<flag>,<host>[,<authuser>,<authpasswd>]`
	Response
	If there are more commands need to enter:
	`+HTTP CMD: CONTIUE ENTER CMD`
	If all commands has enter:
	`+HTTPCREATE: <httpclientId>`
	If there is any error, response:
	`+HTTP ERROR: <err>`

Test Command `AT+HTTPCREATE=?`	Response `+HTTPCREATE:` (list of supported< flag >s), "<host>" , "<authuser>" , "<authpasswd>" OK
Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

Parameter

<flag>	Integer type	
	1	not last part of command
	0	last part of command
<host>	string type	
	http server's host name	

<authuser>	String type Authentication user name
<authpasswd>	String type Authentication password
< httpclientId >	Integer type http Client Id ,0

Example

```
AT+HTTPCREATE=0,"http://api.openweathermap.org:80"
+HTTPCREATE: 0
OK
```

2.2.2 AT+HTTPCON

Set command creates a socket and connects with a http server. Then creates a task to receive data come from http server.
Test command returns values supported as a compound value.

AT+HTTPCON

Set Command AT+HTTPCON=<httpclientId>	Response OK If there is any error, response: +HTTP ERROR: <err>
AT+HTTPCON=?	Response +HTTPCON: (list of supported< httpclientId >) OK
Maximum Response Time	40s
Parameter Saving Mode	NO_SAVE

Parameter

<httpclientId>	Integer type http client id returned by +HTTPCREATE

Example

```
AT+HTTPCON=0
OK
```

2.2.3 AT+HTTPDESTROY

Set command closes a socket, stops receive data from the http server and free the memory that was allocated by the client when creation.
Test command returns values supported as a compound value.

AT+HTTPDESTROY

Set Command AT+HTTPDESTROY=<httpclientId>	Response OK If there is any error, response: +HTTP ERROR: <err>
AT+HTTPDESTROY=?	Response +HTTPDESTROY: (list of supported< httpclientId >) OK
Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

Parameter

<httpclientId>	Integer type
	http client id returned by +HTTPCREATE

Example

```
AT+HTTPDESTROY=0
OK
```

2.2.4 AT+HTTPSEND

Set command sends data to the http server.
Test command returns values supported as a compound value.
NOTE: only one send command could be processing before the related receiving is complete.
e.g. 2nd AT+HTTPSEND=xxx will return +HTTP ERROR: SEND FAILED.

AT+HTTPSEND

Set Command AT+HTTPSEND=<httpclientId>,<method>,<pathlen>,<path>,<customheaderlen>,<customheader>,<contentTypelen>,<contentType>,<contentlen>,<content>	Response OK If there is any error, response: +HTTP ERROR: <err>
AT+HTTPSEND=?	Response +HTTPSEND: (list of supported< httpclientId>),(list of supported< method>),(range of supported< pathlen>), “<path>” ,(range of supported< customheaderlen>), “< customheader>” ,(range of supported< contentTypelen>), “< contentType>” ,(range of supported< contentlen>), “<content>” OK

Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

Parameter

<httpclientId>	Integer type
	http client id returned by +HTTPCREATE
<method>	Integer type; http method
	0 GET
	1 POST
	2 PUT
	3 DELETE
	4 HEAD
<pathlen>	Integer type
	Lenth of path,0-260
<path>	string type
	Path
<customheaderlen>	Integer type
	Length of custom header,0-255
<customheader>	string type
	Customheader in hex string
<contentTypelen>	Integer type
	Length of content type,0-64
<contentType>	string type
	Content type
<contentlen>	Integer type,0-1024
	Length of content
<content>	String type
	User data need to send in hex string

Example

```
AT+HTTPSEND=0,0,89, "/data2.5/weather?q=shanghai&
appid=c592e14137c3471fa9627b44f6649db4&mode=xml&units=metric”
OK
```

2.2.5 +HTTPRESPH indicator of response header

This is an unsolicited message to represent response header.

+HTTPRESPH

```
+HTTPRESPH: <clientId>,<responseCode>,<headerlen>,<header>
```

Parameter

<clientId>	Integer type	
	http client id returned by +HTTPCREATE	
<responseCode>	Integer type	
	http response code	
<headerlen>	Integer type	
	Length of http response header	
<header>	string type	
	Header	

2.2.6 +HTTPRESPC indicator of response content

This is an unsolicited message to represent response content.

+HTTPRESPC

```
+HTTPRESPC: <clientId>,<flag>,<contentlength>,<blockcontentlen>,<content>
```

Parameter

<clientId>	Integer type	
	http client id returned by +HTTPCREATE	
<flag>	Integer type; if has more data	
	0	No more data
	1	Has more data
<contentlength>	Integer type	
	Length of content	
<blockcontentlen>	Integer type	
	Current block length	
<content>	string type	
	content data string which is converted from content hex data, the length is 2*original hex data.	

2.2.7 +HTTPERR indicator of error message

This is an unsolicited message to represent error message when error happen.

+HTTPERR

```
+HTTPERR: <clientId>,<errorcode>, [<rspcode>]
```

Parameter

<clientId>	Integer type
	http client id returned by +HTTPCREATE
<errorcode>	Integer type

2	URL parse error
3	could not resolve DNS
4	Protocol error
5	HTTP 404 error, NOT FOUND
6	HTTP 403 error, REFUSED
7	HTTP xxx error, with specific rspcode followed
8	Connection timeout
9	Connection error
10	Connection fatal error
11	Connection closed
13	Buffer overflow error

3. Appendix

3.1 Power-on inspection

（1）AT // Determine if the module is powered on and started successfully

（2）AT+CFUN=1 // Turn off airplane mode

（3）AT+CEREG? // determine the PS domain attachment status. The second parameter is 1 or 5 indicating normal attachment

3.2 Access the HTTP server

（1）AT+HTTPCREATE=0,“http://api.openweathermap.org:80”

（2）AT+HTTPCON=0

（3）AT+HTTPSEND=0,0,90,"/data/2.5/weather?q=shanghai&
appid=c592e14137c3471fa9627b44f6649db4&mode=xml&units=metric"

4. Version

版本	日期	备注
Draft	2018-4-2	
A		

5. About US

Shanghai EigenCOMM Technology Co.,Ltd is established in Feb 2017 in Zhangjiang Hi-Tech park, Shanghai China (www.eigencomm.com) . EigenCOMM focuses on cellular based IoT chipset and solution. With accumulation of experience over the years in algorithm, L1/L2/L3 protocol, SoC, RF , platform & application software as well as the communication architecture and system, team makes its way to IoT industry.

The ultra-low power NB-IoT chipset EC616(S) has already enter mass production stage. The 4G Cat.1 chipset EC618 is in engineering samples stage, 5G is also in the R&D stage.

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